

Models to simulate real-world scenarios

Module III Simulation Scenarios for IoT data

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SCOPE,VIT

- Machine Learning
- Regression Methods
- Supervised ML
- Unsupervised ML
- Artificial Intelligence

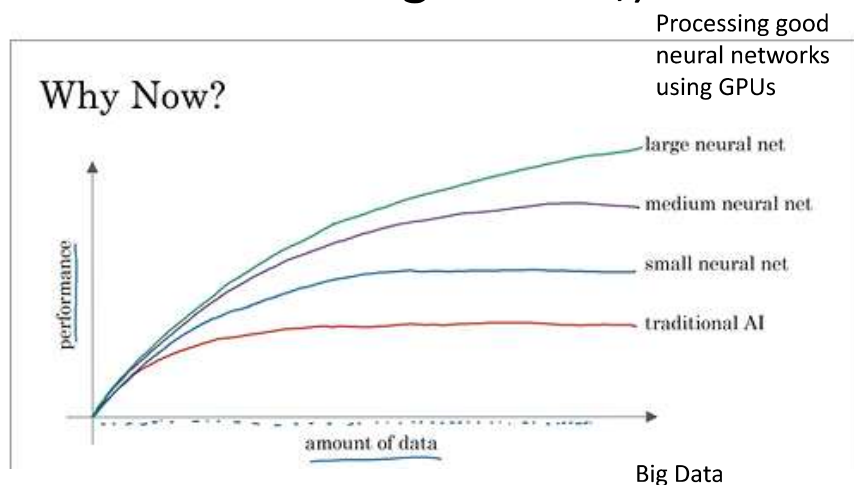
Terminologies

- Machine learning
- Data science

Machine Learning

Input (A) →	Output (B)	Application
email →	spam? (0/1)	spam filtering
audio →	text transcript	speech recognition
English →	Chinese	machine translation
ad, user info →	click? (0/1)	online advertising
image, radar info →	position of other cars	self-driving car
image of phone →	defect? (0/1)	visual inspection

Importance of data in AI & ML (consider speech recognition, self driving car etc.,)



Importance of Data set

size of house (square feet)	# of bedrooms	price (1000\$)
523	1	115
645	1	150
708	2	210
1034	3	280
2290	4	355
2545	4	440

image	label
	cat
	not cat
	cat
	not cat

Acquiring data

- Manual labelling



- From observing behaviours

user ID	time	price (\$)	purchased
4783	Jan 21 08:15:20	7.95	yes
3893	March 3 11:30:15	10.00	yes
8384	June 11 14:15:05	9.50	no
0931	Aug 2 20:30:55	12.90	yes

machine	temperature (°C)	pressure (psi)	machine fault
17987	60	7.65	N
34672	100	25.50	N
08542	140	75.50	Y
98536	165	125.00	Y

Data is messy

- Garbage in, garbage out
- Data problems
- Incorrect labels
- Missing values

size of house (square feet)	# of bedrooms	price (1000\$)
523	1	115
645	1	0.001
708	unknown	210
1034	3	unknown
unknown	4	355
2545	unknown	440

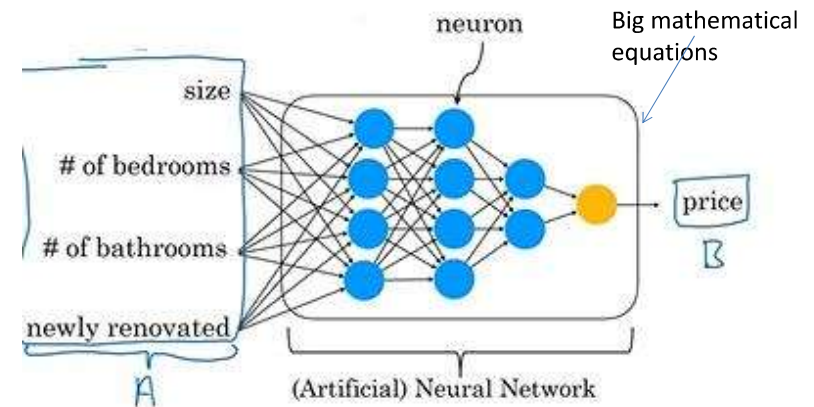
- Multiple types of data
- Unstructured data
 - Image, audio, text

Structured data

Data science

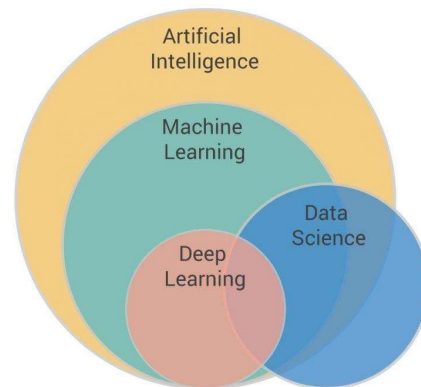
- Science of extracting knowledge and insights from data.
- Analysing the results and bringing the conclusions, predictions
 - Home with 3 bedrooms are more expensive than homes with 2 bedrooms of a similar size.
 - Newly renovated homes have a 15% premium.
- Online advertising
 - continuously running the advertising software and provide the sales department about the analysis of products to run their business

Deep Learning



AI tools

- Machine learning and data science
- Deep learning/ neural networks
- Other tools:
 - Unsupervised learning, reinforcement learning, graphical models, planning, knowledge graph



Machine Learning Methods (Applications of models)

1. Regression Methods

Three types of data

- Linear data
 - Regression methods
- Labelled data
 - Supervised machine learning methods
- Unlabelled data
 - Unsupervised machine learning methods

Uses of Regression

- Identify the strength of independent variables on dependent variables
 - What is the relationship between the sales and marketing spending
 - What is the relation ship between age and income
- Forecasting an effect
 - How much dependent variables effect if one or more independent variables changes.
 - Evaluating trends and sales estimates
 - Analysing the impact of price changes
 - How much steel I will get for each 100\$ change.
- Trend forecasting
 - For future values
 - What is the value of my Bitcoin in the next two months.
 - Assessment of risk in financial services and insurance domain

What is Regression

- Form of predictive modelling technique to investigate the relation between **independent** and **dependent** variables.
- Graphing a line that is mostly to close to the set of data points.

Types of regression analysis

- Unitary linear regression analysis
 - One independent variable and one independent variable regression analysis. And the approximation relation between the two is straight line.
 - The kind of relationship between the two is straight line.
- Multivariate linear regression analysis
 - Two or more independent variables.
 - The relationship between the two variables is straight line, it is multivariate linear regression analysis.
- Logistic regression analysis

Steps in Linear regression

Observe the data draw the scatter diagram

Fitting by least square method

Get the function $f(x)$ and verify it

Predict unknown data and make conclusion

- Scatter the points as diagram. Find the approximation equation.
- Reduce the random error by using least square method and obtain the expression.
- Fit the expression with the data and check it by using coefficient of determinations R^2 which is close to 1.
- Predicting the problems as true or false

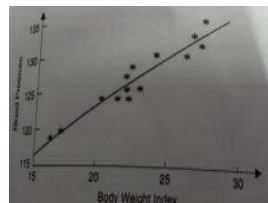
Health care data analysis with linear regression (Example)

- Obesity is reflected by weight index. A fat person have high blood pressure or diabetes.

id	Body Weight Index	Blood Pressure (mmHg)
1	20.5	121
2	21.5	123
3	19.6	123
4	28	130
5	16.5	119
6	25.9	131
7	21.6	125

Procedure

- Determine the distribution of data points.
- Draw the scatter diagram for Body weight and blood pressure using any tool.
- All points are linearly distributed , so can model using unitary linear regression process.
- Perform linear square method to obtain a and b. (a or m = 1.32 and b or c or β =96.58).
 - $Y = 1.32x + 96.58$
- Significant test is performed to check the fitness of the data (obtained R^2 as $0.90 < 1$ and close to 1)
- So the expression is significant (Gradient procedure)



Regression formulation process

- Unknown parameters as β
 - Can be scalar or vector.
- Independent variable as X and dependent variable Y.
- Relation among these as:
 - $Y = f(X, \beta)$.

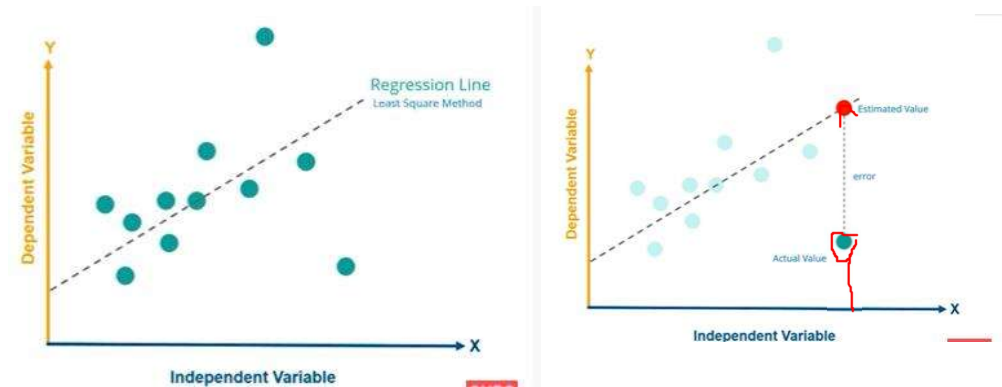
Understanding Linear regression algorithm

- $Y = mx + C$
- Regression line (best line)
- Residual error
 - difference between actual data point to our regression line.
- Goal is to reduce the residual error

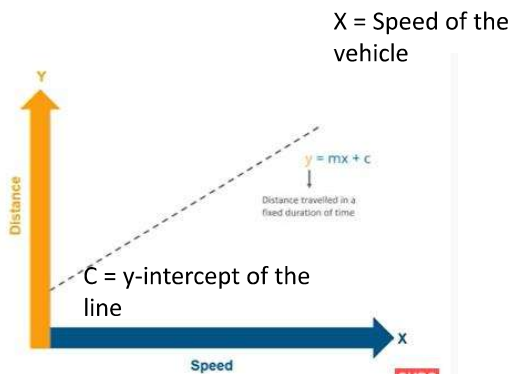
- Positive line regression (m +ve)
- Negative line regression (m -ve)

$y = mx + c$
 $y = mx$
 $y = mx + c$

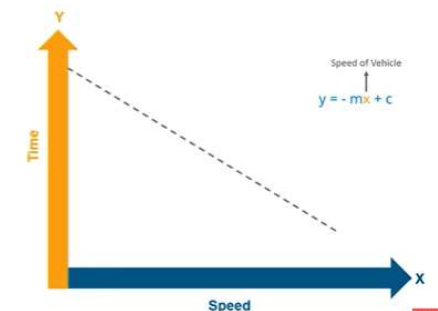
Regression definitions



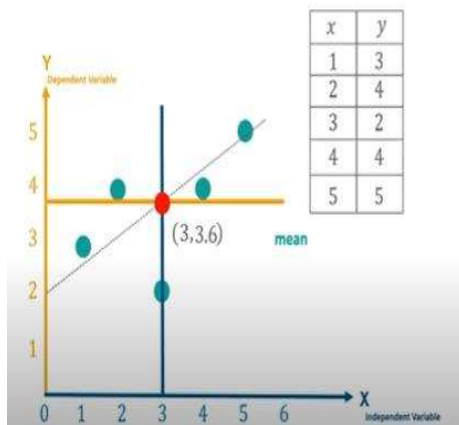
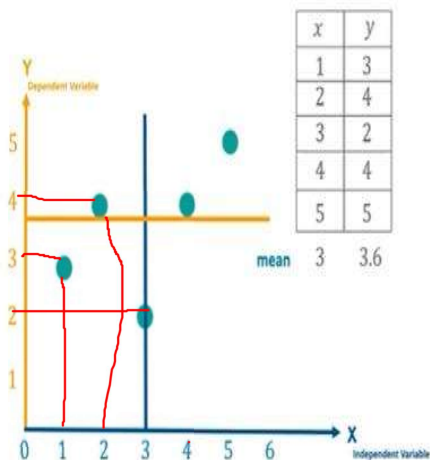
Positive Line regression



Negative Line regression



Example using Linear Regression algorithm



Finding m

- $X \rightarrow$ X data point
- $X' \rightarrow$ mean of X
- $Y \rightarrow$ Y data point
- $Y' \rightarrow$ Mean of Y
- $X - X' \rightarrow$ the distance of data points from Y mean
- $Y - Y' \rightarrow$ the distance of data points from X mean.

$$m = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$

Procedure to obtain linear regression line

x	y
1	3
2	4
3	2
4	4
5	5

mean 3 3.6

Step 1

x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})^2$	$(x - \bar{x})(y - \bar{y})$
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Step 2

$$m = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$

Step 3

Step 4

Substitute any input values like $x = 3$ and $y = 3.6$, based on m, Find C?

Step 5 : Substitute in $Y = mx + c$

x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})^2$	$(x - \bar{x})(y - \bar{y})$
1	3	-2	-0.6	4	1.2
2	4	-1	0.4	1	-0.4
3	2	0	-1.6	0	0
4	4	1	0.4	1	0.4
5	5	2	1.4	4	2.8

mean 3 3.6

x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})^2$	$(x - \bar{x})(y - \bar{y})$
1	3	-2	-0.6	4	1.2
2	4	-1	0.4	1	-0.4
3	2	0	-1.6	0	0
4	4	1	0.4	1	0.4
5	5	2	1.4	4	2.8

mean 3 3.6 $\Sigma = 10$ $\Sigma = 4$

$$m = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2} = \frac{4}{10}$$

- Substitute the values of $x = 3$ and $y = 3.6$, $m = 0.4$

What Is C??

2.4

Equation becomes :

$$y = 0.4x + 2.4$$

$$\square \therefore c = 0.4 \times 3 + c$$

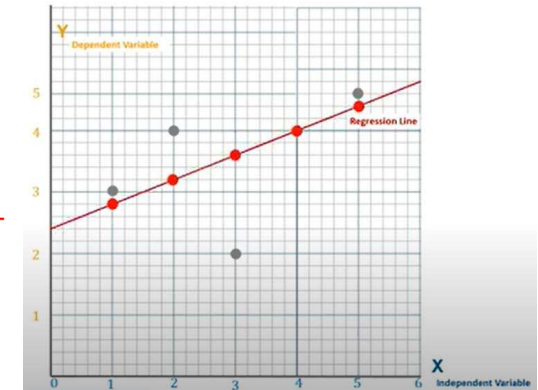
Step 6 : Calculate all Y predicted values

- Find Y(Predicted) :Yp
- Substitute for every Xi, $\rightarrow$ find Yi

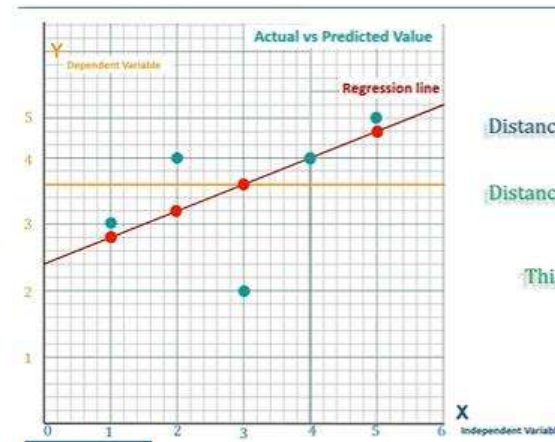
$$m = 0.4$$
$$c = 2.4$$
$$y = 0.4x + 2.4$$

For given $m = 0.4$ & $c = 2.4$, lets predict values for y for $x = \{1, 2, 3, 4, 5\}$

$$y = 0.4 \times 1 + 2.4 = 2.8$$
$$y = 0.4 \times 2 + 2.4 = 3.2$$
$$y = 0.4 \times 3 + 2.4 = 3.6$$
$$y = 0.4 \times 4 + 2.4 = 4.0$$
$$y = 0.4 \times 5 + 2.4 = 4.4$$



Calculation of R^2



Distance actual - mean

vs

Distance predicted - mean

This is nothing but $R^2 = \frac{\sum (y_p - \bar{y})^2}{\sum (y - \bar{y})^2}$

Fitting line

STEP 7 : GRADIENT PROCEDURE TO PERFORM THE ITERATIONS FOR BEST FIT LINE

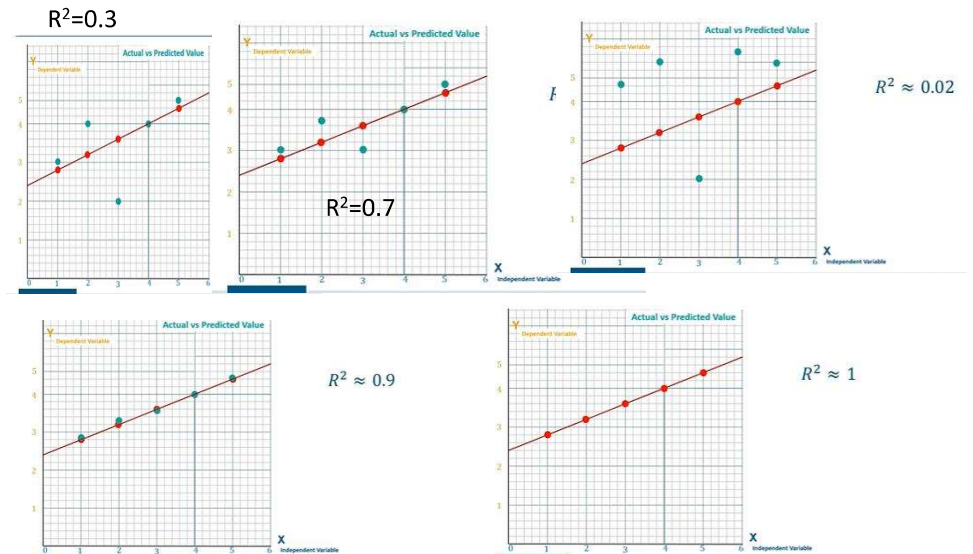
Calculate the R squared value

x	y	y - $\bar{y}$	(y - $\bar{y}$) ²	$\hat{y}_p$	($\hat{y}_p$ - y)	($\hat{y}_p$ - y) ²
1	3	-0.6	0.36	2.8	-0.8	0.64
2	4	0.4	0.16	3.2	-0.4	0.16
3	2	-1.6	2.56	3.6	0	0
4	4	0.4	0.16	4.0	0.4	0.16
5	5	1.4	1.96	4.4	0.8	0.64
mean y	3.6		Σ 5.2		Σ 1.6	

$$R^2 = \frac{1.6}{5.2} = \frac{\Sigma (\hat{y}_p - y)^2}{\Sigma (y - \bar{y})^2}$$

$$R^2 \approx 0.3$$

Variation of data points & R²



Sample Data set (A1)

- Here's a sample **IoT dataset** with 5 data points for solving a **linear regression** equation. This dataset represents temperature sensor readings and their corresponding power consumption in an industrial setting.

Sample IoT Dataset (Temperature vs. Power Consumption)

Temperature (°C)	Power Consumption (kW)
20	200
25	250
30	300
35	350
40	400

Sample Data set (A2)

- Here's a **sample IoT dataset** related to **healthcare**, where we analyze the relationship between **heart rate (bpm)** and **calories burned per hour** using **linear regression**.

Sample Healthcare IoT Dataset (Heart Rate vs. Calories Burned)

Heart Rate (bpm)	Calories Burned (kcal/hour)
60	180
70	210
80	240
90	270
100	300

What is Multiple Linear Regression(MLR)

- It is a statistical technique that uses several explanatory (independent) variables to predict the outcome of a response variable
- the extension of ordinary least-squares (OLS) or simple linear regression because it involves more than one explanatory variable.
- A multiple regression model extends to several explanatory variables.
- It is used to determine a mathematical relationship among a number of random variables

Points to consider MLR

- There is a linear relationship between the dependent variables and the independent variables
- The independent variables are not too highly correlated with each other
- y_i observations are selected independently and randomly from the population
- Residuals(ϵ) should be normally distributed with a mean of 0 and variance σ

Uses of Multiple Linear Regression

- To determine a mathematical relationship among a number of random variables.
- MLR examines how multiple independent variables are related to one dependent variable.
- The model creates a relationship in the form of a straight line (linear) that best approximates all the individual data points.

Formula and calculation of MLR

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip} + \epsilon$$

where, for $i = n$ observations:

y_i = dependent variable

x_i = explanatory variables

β_0 = y-intercept (constant term)

β_p = slope coefficients for each explanatory variable

ϵ = the model's error term (also known as the residuals)

Coefficient of determination (R-Squared)

- It is a statistical metric that is used to measure how much of the variation in outcome can be explained by the variation in the independent variables.
- R^2 always increases as more predictors are added to the MLR model, even though the predictors may not be related to the outcome variable.
- R^2 can only be between 0 and 1, where 0 indicates that the outcome cannot be predicted by any of the independent variables and 1 indicates that the outcome can be predicted without error from the independent variables

MLR Equation

- y_i = dependent variable—the price of XOM
- x_{i1} = interest rates
- x_{i2} = oil price
- x_{i3} = value of S&P 500 index
- x_{i4} = price of oil futures
- B_0 = y-intercept at time zero
- B_1 = [regression coefficient](#) that measures a unit change in the dependent variable when x_{i1} changes - the change in XOM price when interest rates change
- B_2 = coefficient value that measures a unit change in the dependent variable when x_{i2} changes—the change in XOM price when oil prices change

Example

- Problem statement :how the movement of the market affects the price of ExxonMobile (XOM).
 - linear equation will have the value of the S&P 500 index as the independent variable, or predictor, and the price of XOM as the dependent variable and multiple factors that predict the outcome of an event such as the price of oil, interest rates, and the price movement of oil etc.,

Measurement of coefficient measures

- The least-squares estimates, $B_0, B_1, B_2 \dots B_p$, are usually computed by statistical software. As many variables can be included in the regression model in which each independent variable is differentiated with a number—1,2, 3, 4...p
- Example of MLR equation

XOM Price = 75 - 1.5 interest rates + 7.8 oil price + 3.2 S&P 500 + 5.7 oil futures

R - Sq = 86.5%

Correlation

- Correlation is a statistic measure of two variables closest degree relation.
- Correlation shows the strength of a relationship between two variables and is expressed numerically by the correlation coefficient
- The correlation coefficient's values range between -1.0 and 1.0
 - Positive correlation:
 - means that the correlation coefficient is exactly 1. This implies that as one security moves, either up or down, the other security moves in lockstep, in the same direction
 - Negative correlation
 - means that two assets move in opposite directions
 - Zero correlation
 - implies no linear relationship at all.

Formula for Correlation

$$r = \frac{\sum(X - \bar{X})(Y - \bar{Y})}{\sqrt{\sum(X - \bar{X})^2} \sqrt{\sum(Y - \bar{Y})^2}}$$

where:

r = Correlation coefficient

$\bar{X}$ = Average of observations of variable X

$\bar{Y}$ = Average of observations of variable Y

Differences between LR and MLR

Linear Regression

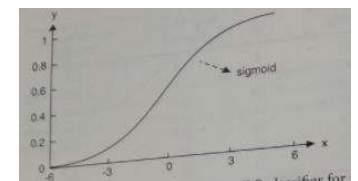
- compares the response of a dependent variable given a change in some explanatory variables

Multiple Linear regression

- Multiple regressions are based on the assumption that there is a linear relationship between both the dependent and independent variables. It also assumes no major correlation between the independent variables.
- Multiple regressions can be linear and nonlinear.

Logistic regression for classification

- It is linear regression in broad sense
- Used for prediction and classification
 - Used for datamining, automatic diagnosis for diseases.
 - Expression and domain is (-infi to + infi) and value is ranging between 0 to 1. so why divided into two classes.
- $X = (X_1, X_2, X_3, \dots, X_m)$
 - X is vector of m features
 - $X_1 \rightarrow$ one attribute of sample data (training data)
 - All the samples are combined into one feature using the linear function



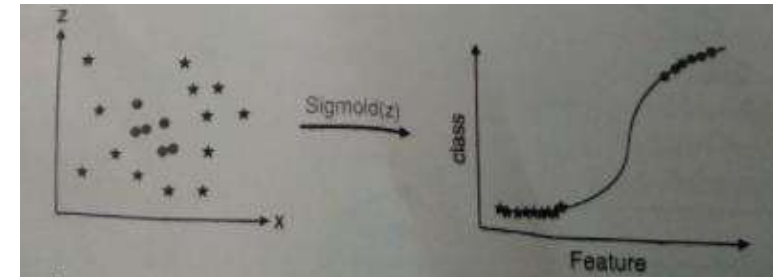
$$f(x) = \frac{1}{1 + e^{-z}}$$

$$z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_m x_m$$

How to solve it using Logistic model

Age	Gender	Resting Heart Rate	Maximum Heart Rate	Heart Rate Cat
45	Male	70	180	High
52	Female	65	175	High
30	Male	80	190	High
62	Female	72	168	High
55	Male	75	160	Moderate
40	Female	70	150	Moderate
48	Male	78	170	Moderate
50	Female	68	155	Moderate
35	Male	85	180	High
42	Female	67	165	Moderate

Principle used for logistic regression for classification



$$\begin{cases} P(Y=1|x) = \pi(x) = \frac{1}{1+e^{-z}} \\ z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_m x_m \end{cases} \rightarrow \begin{cases} x \in 1, \text{ if } P(Y=1|x) > 0.5 \\ x \in 0, \text{ if } P(Y=0|x) < 0.5 \end{cases}$$

How do you label the data as High or Moderate

Age	Gender	Resting Heart Rate	Maximum Heart Rate	
45	Male	70	180	
52	Female	65	175	
30	Male	80	190	
62	Female	72	168	
55	Male	75	160	
40	Female	70	150	
48	Male	78	170	
50	Female	68	155	
35	Male	85	180	
42	Female	67	165	

Machine Learning Methods (Applications of models)

2. Supervised machine learning with Decision tree algorithm

Supervised ML

- Mapping between two entities.
 - $A \rightarrow B$
 - Input to Output (Labelled data)

Classification (Feature of Supervised ML)

- Processing of dividing the datasets into different classes or groups and added with labels.
 - Technique of categorizing various observations into classes or groups.
 - Taking data, analysing it, based on conditions into categorizing
- Why categorization
 - To perform predictions on data
 - Is this male category or female category
 - Is this category A or B or C
- A machine has to train before going for classification.
- Gmail
 - Spam or not a spam

Uses of classification

- To check the transaction is liable or not: Machine predicts about fishing in the credit card transactions.
 - Credit card used in INDIA. Flied to Dubai. Gets an alert for transaction to proceed. Fishing in transactions will be verified (legal / non legal).
- To know about weather category
- To know about food category

Type of classification

- To train the data for classification various model.
 - Decision Tree
 - Random Forest
 - Naive Bayes
 - KNN

Decision Tree

Decision Tree

- Graphical representation of all the possible solutions to a decision
- Decisions are based on some conditions
- Decision made can be easily explained



Deciding to go for restaurant or not?

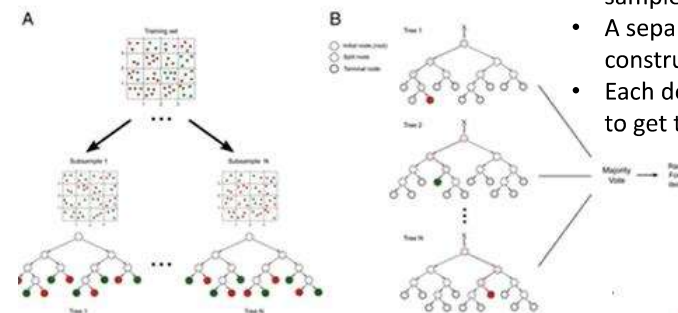
Random Forest

Random Forest

- Builds multiple decision trees and merges them together
- More accurate and stable prediction
- Random decision forests correct for decision trees' habit of overfitting to their training set
- Trained with the "bagging" method

A polling system prediction

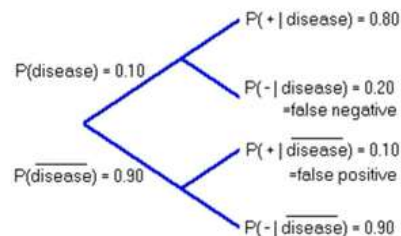
- You have huge data set.
- The data is divided into small sample set
- A separate decision tree is constructed for each sample set.
- Each decision tree is combined to get the final decision.



Naïve Bayes

Naïve Bayes

- Classification technique based on Bayes' Theorem
- Assumes that the presence of a particular feature in a class is unrelated to the presence of any other feature

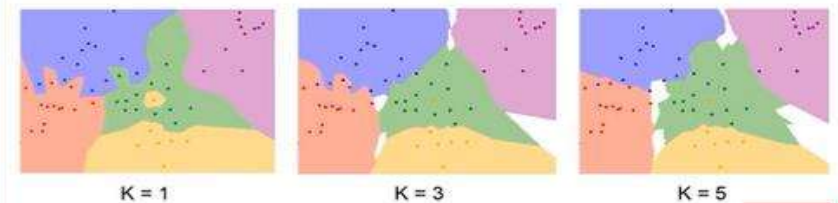


The **Naïve Bayes Theorem** is a **probabilistic classification algorithm** based on **Bayes' Theorem**. It is widely used for **spam filtering, sentiment analysis, and medical diagnosis**. Simple and easy to implement. Used for complex models when the data set is not large. Document classification. In a given text checking the category. In a giving text checking whether a given keyword is present or not. Example : From the given text need to understand whether to understand the probability of disease.

KNN

K-Nearest Neighbors

- Stores all the available cases and classifies new cases based on a similarity measure
- The "K" is KNN algorithm is the nearest neighbors we wish to take vote from.



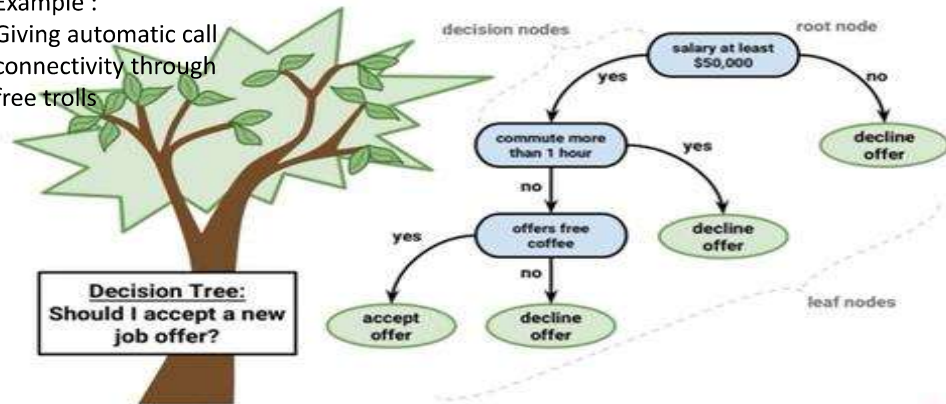
Decision Tree

"A decision tree is a graphical representation of all the possible solutions to a decision based on certain conditions"

Call free dealings
implemented through
Decision tree methods

Example :

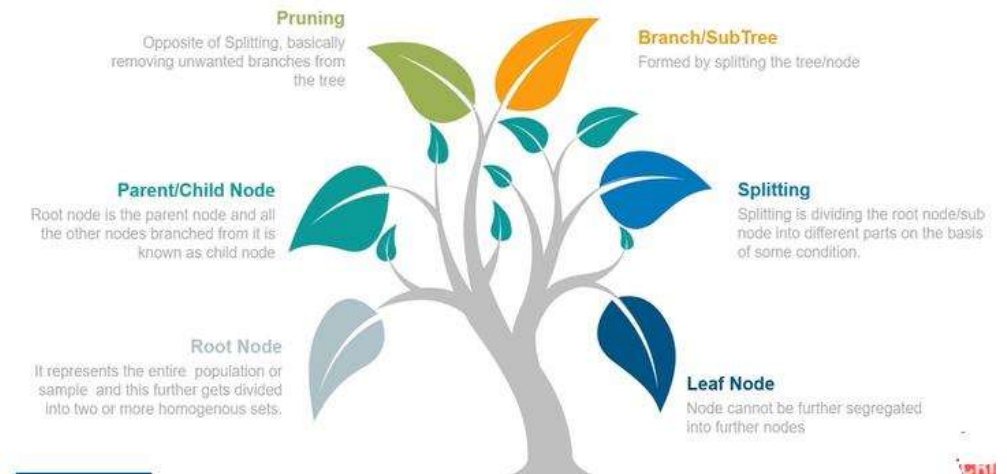
Giving automatic call
connectivity through
free trolls



Example-I

Colour	Diameter	Label
Green	3	Mango
Yellow	3	Mango
Red	1	Grape
Red	1	Grape
Yellow	3	Lemon

Decision Tree Terminology



Deciding a tree how to split

Gini Index
The measure of impurity (or purity) used in building decision tree in CART is Gini Index

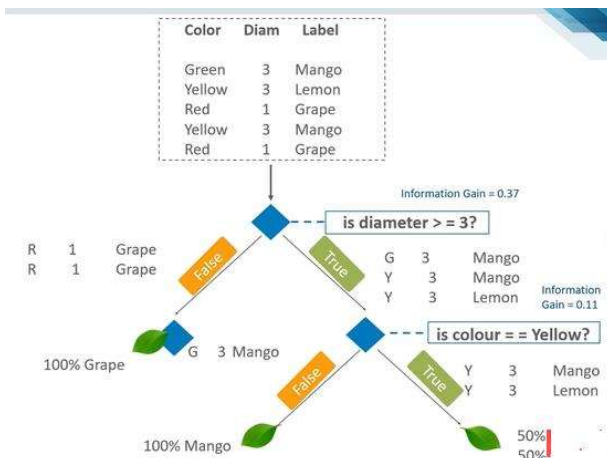
Chi Square
It is an algorithm to find out the statistical significance between the differences between sub-nodes and parent node



Information Gain
The Information gain is the decrease in entropy after a dataset is split on the basis of an attribute. Constructing a decision tree is all about finding attribute that returns the highest information gain

Reduction in Variance
Reduction in variance is an algorithm used for continuous target variables (regression problems). The split with lower variance is selected as the criteria to split the population

CART(Classification And Regression Tree) Algorithm



CART

- Helps in constructing the decision tree

Gini Index

- What questions to which level
- The quantity measure of questions based on Gini Impurity

Steps

- Step 1: Prepare the Dataset
- **Step 2: Calculate the Gini Index**
 - The **Gini Index** measures the impurity of a node. It is calculated as:
 - $Gini(D) = 1 - \sum p_i^2$
 - Where: P_i is the proportion of instances belonging to class i .
- Step 3: Compute Gini Index for Each Feature
 1. **Split the dataset** into subsets based on unique feature values.
 2. **Calculate the weighted Gini Index** after the split:
 - $Gini_{split} = \sum (N_k/N * Gini(D_k))$ Where:
 - N_k is the number of instances in subset k .
 - N is the total number of instances.
 - $Gini(D_k)$ is the Gini Index of subset k .

Decision Tree Using Gini Index

Steps

- **Step 4: Select the Best Feature for Splitting**
 - Choose the feature that results in the **lowest** Gini Index after the split.
 - If the decrease in Gini Index is significant, the split is useful.
- Step 5: Repeat for Each Node
- **Step 6: Construct the Final Decision Tree**
 - The final decision tree consists of **root, internal, and leaf nodes**.
 - Class labels are assigned to the leaf nodes.

Gini Index based on Decision attribute

Weekend	Weather	Parents	Money	Decision
W1	Sunny	Yes	Rich	Cinema
W2	Sunny	No	Rich	Tennis
W3	Windy	Yes	Rich	Cinema
W4	Rainy	Yes	Poor	Cinema
W5	Rainy	No	Rich	Stay In
W6	Rainy	Yes	Poor	Cinema
W7	Windy	No	Poor	Cinema
W8	Windy	No	Rich	Shopping
W9	Windy	Yes	Rich	Cinema
W10	Sunny	No	Rich	Tennis

- Compute the **Gini Index** for the overall collection of training examples.
- There are **four possible output variables** **Cinema, Tennis, Stay In** and **Shopping**.
- The data has **6 instances of Cinema**, **2 instances of Tennis**, **1 instance of Stay In** and **1 of shopping**.

$$Gini(S) = 1 - \left[\left(\frac{6}{10} \right)^2 + \left(\frac{2}{10} \right)^2 + \left(\frac{1}{10} \right)^2 + \left(\frac{1}{10} \right)^2 \right] = 0.58$$

Gini Index based on Money attribute

Weekend	Weather	Parents	Money	Decision
W1	Sunny	Yes	Rich	Cinema
W2	Sunny	No	Rich	Tennis
W3	Windy	Yes	Rich	Cinema
W4	Rainy	Yes	Poor	Cinema
W5	Rainy	No	Rich	Stay In
W6	Rainy	Yes	Poor	Cinema
W7	Windy	No	Poor	Cinema
W8	Windy	No	Rich	Shopping
W9	Windy	Yes	Rich	Cinema
W10	Sunny	No	Rich	Tennis

- Computation of **Gini Index for Money** Attribute
- It has **two possible values of Rich (7 examples)** and **Poor (3 examples)**.
- For **Money = Poor**, there are **3 examples with "Cinema"**.
- $Gini(S) = 1 - \left[\left(\frac{3}{3} \right)^2 \right] = 0$
- For **Money = Rich**, there are **2 examples with "Tennis"**, **3 examples with "Cinema"** and **1 example with "Stay in", "Shopping" each**
- $Gini(S) = 1 - \left[\left(\frac{2}{7} \right)^2 + \left(\frac{3}{7} \right)^2 + \left(\frac{1}{7} \right)^2 + \left(\frac{1}{7} \right)^2 \right] = 0.694$
- **Weighted Average(Money)**

$$= 0 * \left(\frac{3}{10} \right) + 0.694 * \left(\frac{7}{10} \right) = 0.486$$

Gini Index based on Parents attribute

Weekend	Weather	Parents	Money	Decision
W1	Sunny	Yes	Rich	Cinema
W2	Sunny	No	Rich	Tennis
W3	Windy	Yes	Rich	Cinema
W4	Rainy	Yes	Poor	Cinema
W5	Rainy	No	Rich	Stay In
W6	Rainy	Yes	Poor	Cinema
W7	Windy	No	Poor	Cinema
W8	Windy	No	Rich	Shopping
W9	Windy	Yes	Rich	Cinema
W10	Sunny	No	Rich	Tennis

- Computation of **Gini Index for Parents** Attribute
- It has two possible values of **Yes (5 examples)** and **No (5 examples)**.
- For **Parents = Yes**, there are **5 examples**, all with **"Cinema"**.
- $Gini(S) = 1 - \left[\left(\frac{5}{5} \right)^2 \right] = 0$
- For **Parents = No**, there are **2 examples with "Tennis"**, **1 example with "Stay in", "Shopping" and "Cinema" each**

$$Gini(S) = 1 - \left[\left(\frac{2}{5} \right)^2 + \left(\frac{1}{5} \right)^2 + \left(\frac{1}{5} \right)^2 + \left(\frac{1}{5} \right)^2 \right] = 0.72$$

$$\text{Weighted Average(Parents)} = 0 * \left(\frac{5}{10} \right) + 0.72 * \left(\frac{5}{10} \right) = 0.36$$

Gini Index on Weather

Weekend	Weather	Parents	Money	Decision
W1	Sunny	Yes	Rich	Cinema
W2	Sunny	No	Rich	Tennis
W3	Windy	Yes	Rich	Cinema
W4	Rainy	Yes	Poor	Cinema
W5	Rainy	No	Rich	Stay In
W6	Rainy	Yes	Poor	Cinema
W7	Windy	No	Poor	Cinema
W8	Windy	No	Rich	Shopping
W9	Windy	Yes	Rich	Cinema
W10	Sunny	No	Rich	Tennis

- Computation of **Gini Index for Weather** Attribute
- It has three possible values of **Sunny (3 examples)**, **Rainy (3 examples)** and **Windy (4 examples)**.
- For **Weather = Sunny**, there are **2 examples with "Cinema"** and **1 with "Tennis"**.
- $Gini(Sunny) = 1 - \left[\left(\frac{2}{3} \right)^2 + \left(\frac{1}{3} \right)^2 \right] = 0.444$
- For **Weather = Rainy**, there are **2 examples with "Cinema"** and **1 example with "Stay in"**
- $Gini(Rainy) = 1 - \left[\left(\frac{2}{3} \right)^2 + \left(\frac{1}{3} \right)^2 \right] = 0.444$
- For **Weather = Windy**, there are **3 examples with "Cinema"** and **1 example with "Shopping"**
- $Gini(Windy) = 1 - \left[\left(\frac{3}{4} \right)^2 + \left(\frac{1}{4} \right)^2 \right] = 0.375$

Insights

After 1st level

Weekend	Weather	Parents	Money	Decision
W1	Sunny	Yes	Rich	Cinema
W2	Sunny	No	Rich	Tennis
W3	Windy	Yes	Rich	Cinema
W4	Rainy	Yes	Poor	Cinema
W5	Rainy	No	Rich	Stay In
W6	Rainy	Yes	Poor	Cinema
W7	Windy	No	Poor	Cinema
W8	Windy	No	Rich	Shopping
W9	Windy	Yes	Rich	Cinema
W10	Sunny	No	Rich	Tennis

Weighted Average(Weather)

$$= 0.444 * \left(\frac{3}{10}\right) + 0.444 * \left(\frac{3}{10}\right) + 0.375 * \left(\frac{4}{10}\right)$$

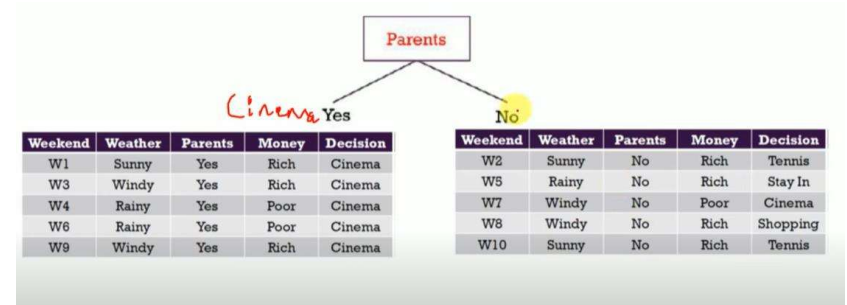
$$= 0.416$$

For Weather - Gini Index: 0.416

For Parents - Gini Index: 0.36

For Money - Gini Index: 0.486

Parents is selected as it has smallest Gini index.



Weekend	Weather	Parents	Money	Decision
W2	Sunny	No	Rich	Tennis
W5	Rainy	No	Rich	Stay In
W7	Windy	No	Poor	Cinema
W8	Windy	No	Rich	Shopping
W10	Sunny	No	Rich	Tennis

Computation of Gini Index for Parents = No | Weather Attribute

- Windy (2 example)
- For Parents = No | Weather = Windy, there is 1 example with "Cinema" and 1 example with "Shopping".
- $Gini(S) = 1 - \left[\left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2\right] = 0.5$

Weighted Average(Parents = No | We

Computation of Gini Index for Parents = No

- Rich (4 examples)
- For Parents = No | Money = Rich, there is each and 2 examples of "Tennis".
- $Gini(S) = 1 - \left[\left(\frac{1}{4}\right)^2 + \left(\frac{1}{4}\right)^2 + \left(\frac{2}{4}\right)^2\right] =$

Computation of Gini Index for Parents = No | Money Attribute

- Poor (1 example)
- For Parents = No | Money = Poor, there is 1 example with "Cinema".
- $Gini(S) = 1 - \left[\left(\frac{1}{1}\right)^2\right] = 0$
- Weighted Average(Parents = No | Money) = $0.625 * (4/5) + 0 * (1/5) = 0.5$

Weekend	Weather	Parents	Money	Decision
W2	Sunny	No	Rich	Tennis
W5	Rainy	No	Rich	Stay In
W7	Windy	No	Poor	Cinema
W8	Windy	No	Rich	Shopping
W10	Sunny	No	Rich	Tennis

For Parents = No | Weather - Gini Index: 0.2

For Parents = No | Money - Gini Index: 0.5

Weather is selected as it has smallest Gini index.

Weekend	Weather	Parents	Money	Decision
W2	Sunny	No	Rich	Tennis
W5	Rainy	No	Rich	Stay In
W7	Windy	No	Poor	Cinema
W8	Windy	No	Rich	Shopping
W10	Sunny	No	Rich	Tennis

Now, for Parent=No & Weather=Sunny, we have all instances as Tennis.

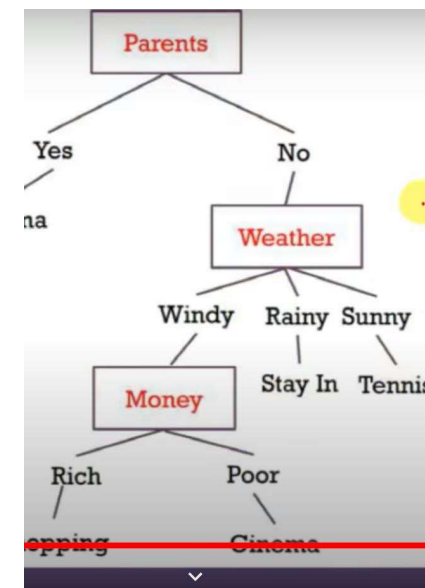
Weekend	Weather	Parents	Money	Decision
W2	Sunny	No	Rich	Tennis ✓
W10	Sunny	No	Rich	Tennis ✓

Now, for Parent=No & Weather=Windy, we need to split.

Now, for Parents=No & Weather=Rainy, we have all instances as Stay In.

Weekend	Weather	Parents	Money	Decision
W5	Rainy	No	Rich	Stay In ✓

Weekend	Weather	Parents	Money	Decision
W7	Windy	No	Poor	Cinema
W8	Windy	No	Rich	Shopping



Construction of Decision Tree

Using ID3 algorithm

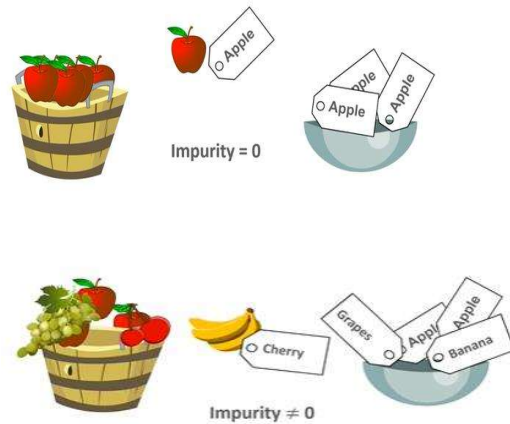
Uses Entropy & Information Gain

Data set for playing a cricket

Day	Outlook	Temp	Humidity	Wind	Play Tennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Impurity

- Degree of randomness:
- How random a data is
 - How the item is correctly mapped with correct label
 - Probability of picking data item to correct label



First step

ENTROPY : HOW TO DECIDE BEST ATTRIBUTE , ABOUT THE IMPURITY OF THE DATA

Entropy

- About the measure of impurity

$$\text{Entropy}(s) = -P(\text{yes}) \log_2 P(\text{yes}) - P(\text{no}) \log_2 P(\text{no})$$

Where,

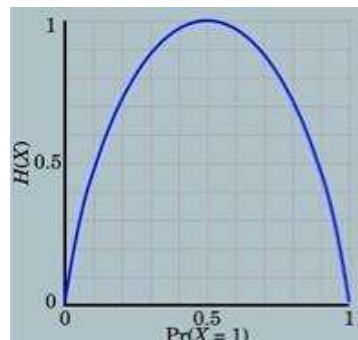
- S is the total sample space,
- $P(\text{yes})$ is probability of yes

If number of yes = number of no ie $P(S) = 0.5$

$$\Rightarrow \text{Entropy}(s) = 1$$

If it contains all yes or all no ie $P(S) = 1$ or 0

$$\Rightarrow \text{Entropy}(s) = 0$$



$$E(S) = -P(\text{Yes}) \log_2 P(\text{Yes})$$

When $P(\text{Yes}) = P(\text{No}) = 0.5$ ie $\text{YES} + \text{NO} = \text{Total Sample}(S)$

$$E(S) = 0.5 \log_2 0.5 - 0.5 \log_2 0.5$$

$$E(S) = 0.5(\log_2 0.5 - \log_2 0.5)$$

$$E(S) = 1$$

$$E(S) = -P(\text{Yes}) \log_2 P(\text{Yes})$$

When $P(\text{Yes}) = 1$ ie $\text{YES} = \text{Total Sample}(S)$

$$E(S) = 1 \log_2 1$$

$$E(S) = 0$$

$$P(\text{no}) \log_2 P(\text{no}) = 0$$

$$E(S) = -P(\text{No}) \log_2 P(\text{No})$$

When $P(\text{No}) = 1$ ie $\text{No} = \text{Total Sam}$

$$E(S) = 1 \log_2 1$$

$$E(S) = 0$$

$$P(\text{yes}) \log_2 P(\text{yes}) = 0$$

Information Gain

- Measures the reduction in entropy
- Decides which attribute should be selected as the decision node

Reference : <https://medium.com/@naeemsunesara/giniscare-entropy-and-information-gain-in-decision-trees-cbc08589852d>

INFORMATION GAIN

If S is our total collection,

Information Gain = Entropy(S) - [(Weighted Avg) x Entropy(each feature)]

Steps to calculate

- Step 1: Prepare the Dataset
- Step 2: Calculate Entropy (H)
 - Entropy measures impurity in a dataset and is calculated as:
 - $H(D) = -\sum p_i \log_2 p_i$ Where:
 - p_i is the proportion of class i in the dataset.
- Step 3: Compute Entropy for Each Feature Split For each feature:
 - Split the dataset based on unique values or thresholds.
 - Calculate the weighted entropy after the split:
 - $H_{split} = \sum (N_k/N) * H(D_k)$
 - Where: N_k is the number of instances in subset k .
 - N is the total number of instances.
 - $H(D_k)$ is the entropy of subset k .
- Step 4: Calculate Information Gain (IG)
 - Information Gain is the reduction in entropy after splitting:
 - $IG = H(D) - H_{split}$.
- Step 5: Choose the Best Feature for Splitting
 - Select the feature with the highest Information Gain.
 - This feature becomes the decision node of the tree.
- Step 6: Repeat for Each Node
 - Apply Steps 2-5 recursively to child nodes.
 - Stop when:
 - The node is pure (all samples belong to one class).
 - A stopping criterion (e.g., max depth, min samples per node) is reached.
- Step 7: Construct the Final Decision Tree
 - The root node is the feature with the highest Information Gain.
 - Branches represent feature values leading to child nodes.
 - Leaf nodes contain class labels.

Compute the entropy for the Data set

Out of 14 instances we have 9 YES and 5 NO

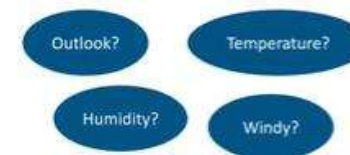
So we have the formula,

$$E(S) = -P(\text{Yes}) \log_2 P(\text{Yes}) - P(\text{No}) \log_2 P(\text{No})$$

$$E(S) = -(9/14) * \log_2 9/14 - (5/14) * \log_2 5/14$$

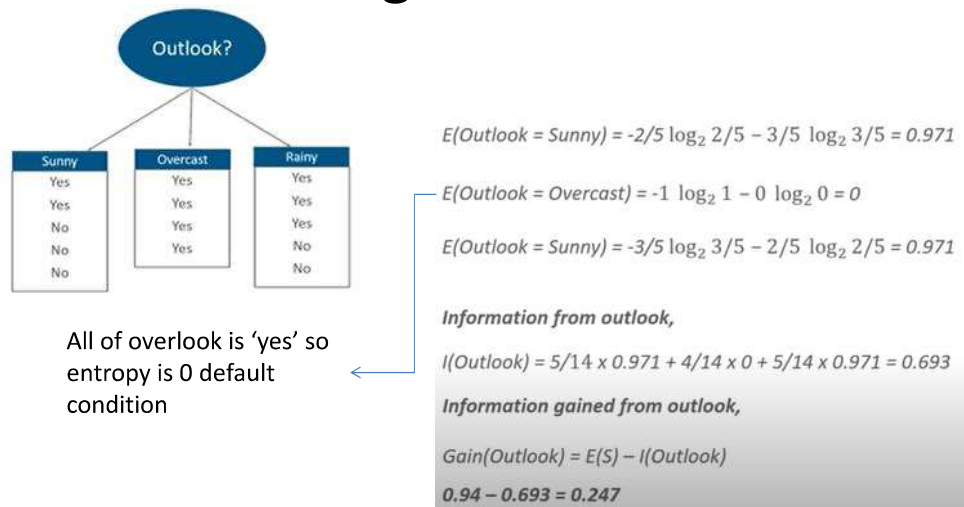
$$E(S) = 0.41 + 0.53 = 0.94$$

Which attribute as root



Day	Outlook	Temp	Humidity	Wind	Play Tennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Information gained from Outlook



Information gained from windy



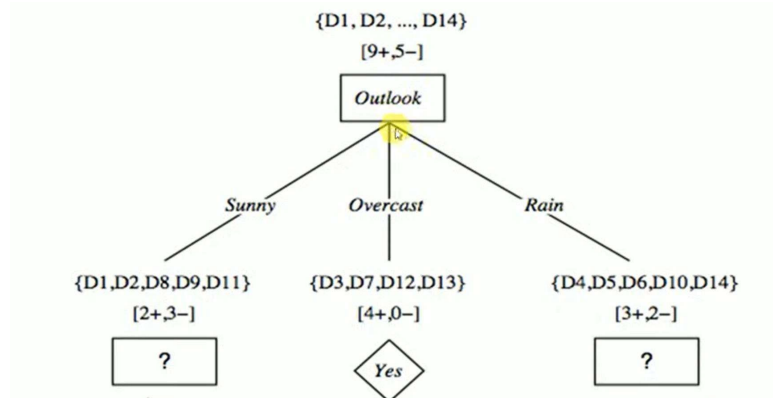
Information gained from all

Outlook:		Temperature:	
Info	0.693	Info	0.911
Gain: 0.940-0.693	0.247	Gain: 0.940-0.911	0.029

Humidity:		Windy:	
Info	0.788	Info	0.892
Gain: 0.940-0.788	0.152	Gain: 0.940-0.982	0.048

Since Max gain = 0.247,
Outlook is our ROOT Node

Decision tree after root Node



Information Gain for Temp

Day	Temp	Humidity	Wind	Play Tennis
D1	Hot	High	Weak	No
D2	Hot	High	Strong	No
D8	Mild	High	Weak	No
D9	Cool	Normal	Weak	Yes
D11	Mild	Normal	Strong	Yes

Attribute: Temp

Values (Temp) = Hot, Mild, Cool

$$\begin{aligned}
 S_{Sunny} &= [2+, 3-] & Entropy(S_{Sunny}) &= -\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5} = 0.97 \\
 S_{Hot} &\leftarrow [0+, 2-] & Entropy(S_{Hot}) &= 0.0 \\
 S_{Mild} &\leftarrow [1+, 1-] & Entropy(S_{Mild}) &= 1.0 \\
 S_{Cool} &\leftarrow [1+, 0-] & Entropy(S_{Cool}) &= 0.0
 \end{aligned}$$

$$\begin{aligned}
 Gain(S_{Sunny}, Temp) &= Entropy(S) - \sum_{v \in \{Hot, Mild, Cool\}} \frac{|S_v|}{|S|} Entropy(S_v) \\
 Gain(S_{Sunny}, Temp) &= Entropy(S) - \frac{2}{5} Entropy(S_{Hot}) - \frac{2}{5} Entropy(S_{Mild}) - \frac{1}{5} Entropy(S_{Cool}) \\
 Gain(S_{Sunny}, Temp) &= 0.97 - \frac{2}{5} \cdot 0.0 - \frac{2}{5} \cdot 1 - \frac{1}{5} \cdot 0.0 = 0.570
 \end{aligned}$$

Information Gain for Humidity

Day	Temp	Humidity	Wind	Play Tennis
D1	Hot	High	Weak	No
D2	Hot	High	Strong	No
D8	Mild	High	Weak	No
D9	Cool	Normal	Weak	Yes
D11	Mild	Normal	Strong	Yes

Attribute: Humidity

Values (Humidity) = High, Normal

$$\begin{aligned}
 S_{Sunny} &= [2+, 3-] & Entropy(S) &= -\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5} = 0.97 \\
 S_{High} &\leftarrow [0+, 3-] & Entropy(S_{High}) &= 0.0 \\
 S_{Normal} &\leftarrow [2+, 0-] & Entropy(S_{Normal}) &= 0.0
 \end{aligned}$$

$$\begin{aligned}
 Gain(S_{Sunny}, Humidity) &= Entropy(S) - \sum_{v \in \{High, Normal\}} \frac{|S_v|}{|S|} Entropy(S_v) \\
 Gain(S_{Sunny}, Humidity) &= Entropy(S) - \frac{3}{5} Entropy(S_{High}) - \frac{2}{5} Entropy(S_{Normal}) \\
 Gain(S_{Sunny}, Humidity) &= 0.97 - \frac{3}{5} \cdot 0.0 - \frac{2}{5} \cdot 0.0 = 0.97
 \end{aligned}$$

Information Gain for Wind

Day	Temp	Humidity	Wind	Play Tennis
D1	Hot	High	Weak	No
D2	Hot	High	Strong	No
D8	Mild	High	Weak	No
D9	Cool	Normal	Weak	Yes
D11	Mild	Normal	Strong	Yes

Attribute: Wind

Values (Wind) = Strong, Weak

$$\begin{aligned}
 S_{Sunny} &= [2+, 3-] & Entropy(S) &= -\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5} = 0.97 \\
 S_{Strong} &\leftarrow [1+, 1-] & Entropy(S_{Strong}) &= 1.0 \\
 S_{Weak} &\leftarrow [1+, 2-] & Entropy(S_{Weak}) &= -\frac{1}{3} \log_2 \frac{1}{3} - \frac{2}{3} \log_2 \frac{2}{3} = 0.9183
 \end{aligned}$$

$$\begin{aligned}
 Gain(S_{Sunny}, Wind) &= Entropy(S) - \sum_{v \in \{Strong, Weak\}} \frac{|S_v|}{|S|} Entropy(S_v) \\
 Gain(S_{Sunny}, Wind) &= Entropy(S) - \frac{2}{5} Entropy(S_{Strong}) - \frac{3}{5} Entropy(S_{Weak}) \\
 Gain(S_{Sunny}, Wind) &= 0.97 - \frac{2}{5} \cdot 1.0 - \frac{3}{5} \cdot 0.918 = 0.0192
 \end{aligned}$$

Decision Node

- From all the gains Humidity is having high gain so Humidity is taken as next node.

Day	Temp	Humidity	Wind	Play Tennis
D1	Hot	High	Weak	No
D2	Hot	High	Strong	No
D8	Mild	High	Weak	No
D9	Cool	Normal	Weak	Yes
D11	Mild	Normal	Strong	Yes

$$Gain(S_{Sunny}, Temp) = 0.570$$

$$Gain(S_{Sunny}, Humidity) = 0.97$$

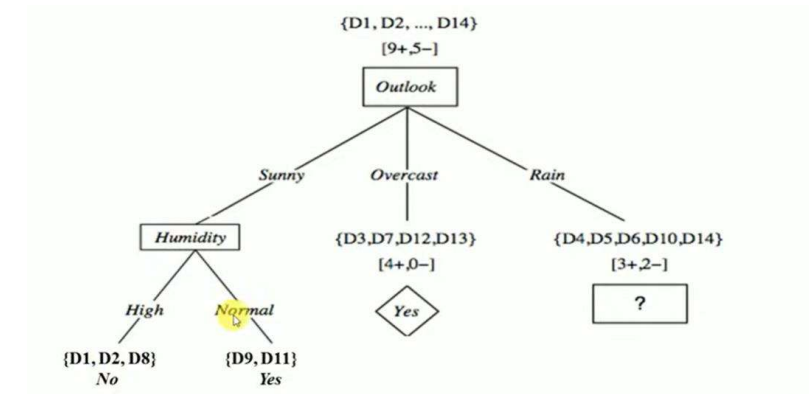
$$Gain(S_{Sunny}, Wind) = 0.0192$$

Conclusion for Level-I (LEFT)

- When taken Humidity has two possibilities like High & Normal
- If Humidity == High the target (Play Tennis) = No & Humidity == LOW the target (play Tennis) = yes

Day	Temp	Humidity	Wind	Play Tennis
D1	Hot	High	Weak	No
D2	Hot	High	Strong	No
D8	Mild	High	Weak	No
D9	Cool	Normal	Weak	Yes
D11	Mild	Normal	Strong	Yes

LEVEL-I Decision tree (LEFT)



Information Gain for Temp

Day	Temp	Humidity	Wind	Play Tennis
D4	Mild	High	Weak	Yes
D5	Cool	Normal	Weak	Yes
D6	Cool	Normal	Strong	No
D10	Mild	Normal	Weak	Yes
D14	Mild	High	Strong	No

Attribute: Temp

Values (Temp) = Hot, Mild, Cool

$$S_{Rain} = [3+, 2-] \quad Entropy(S_{Sunny}) = -\frac{3}{5} \log_2 \frac{3}{5} - \frac{2}{5} \log_2 \frac{2}{5} = 0.97$$

$$S_{Hot} \leftarrow [0+, 0-] \quad Entropy(S_{Hot}) = 0.0$$

$$S_{Mild} \leftarrow [2+, 1-] \quad Entropy(S_{Mild}) = -\frac{2}{3} \log_2 \frac{2}{3} - \frac{1}{3} \log_2 \frac{1}{3} = 0.9183$$

$$S_{Cool} \leftarrow [1+, 1-] \quad Entropy(S_{Cool}) = 1.0$$

$$Gain(S_{Rain}, Temp) = Entropy(S) - \sum_{v \in \{Hot, Mild, Cool\}} \frac{|S_v|}{|S|} Entropy(S_v)$$

$$Gain(S_{Rain}, Temp)$$

$$= Entropy(S) - \frac{0}{5} Entropy(S_{Hot}) - \frac{3}{5} Entropy(S_{Mild})$$

$$- \frac{2}{5} Entropy(S_{Cool})$$

$$Gain(S_{Rain}, Temp) = 0.97 - \frac{0}{5} \cdot 0.0 - \frac{3}{5} \cdot 0.918 - \frac{2}{5} \cdot 1.0 = 0.0192$$

Level-I (construction of Decision Tree for Rainy)

Information Gain for Humidity

Day	Temp	Humidity	Wind	Play Tennis
D4	Mild	High	Weak	Yes
D5	Cool	Normal	Weak	Yes
D6	Cool	Normal	Strong	No
D10	Mild	Normal	Weak	Yes
D14	Mild	High	Strong	No

Attribute: Humidity

Values (Humidity) = High, Normal

$$S_{Rain} = \{3+, 2-\} \quad Entropy(S_{Sunny}) = -\frac{3}{5} \log_2 \frac{3}{5} - \frac{2}{5} \log_2 \frac{2}{5} = 0.97$$

$$S_{High} \leftarrow \{1+, 1-\} \quad Entropy(S_{High}) = 1.0$$

$$S_{Normal} \leftarrow \{2+, 1-\} \quad Entropy(S_{Normal}) = -\frac{2}{3} \log_2 \frac{2}{3} - \frac{1}{3} \log_2 \frac{1}{3} = 0.9183$$

$$Gain(S_{Rain}, Humidity) = Entropy(S) - \sum_{v \in \{High, Normal\}} \frac{|S_v|}{|S|} Entropy(S_v)$$

$$Gain(S_{Rain}, Humidity) = Entropy(S) - \frac{2}{5} Entropy(S_{High}) - \frac{3}{5} Entropy(S_{Normal})$$

$$Gain(S_{Rain}, Humidity) = 0.97 - \frac{2}{5} \cdot 1.0 - \frac{3}{5} \cdot 0.918 = 0.0192$$

Information Gain for Wind

Day	Temp	Humidity	Wind	Play Tennis
D4	Mild	High	Weak	Yes
D5	Cool	Normal	Weak	Yes
D6	Cool	Normal	Strong	No
D10	Mild	Normal	Weak	Yes
D14	Mild	High	Strong	No

Attribute: Wind

Values (wind) = Strong, Weak

$$S_{Rain} = \{3+, 2-\} \quad Entropy(S_{Sunny}) = -\frac{3}{5} \log_2 \frac{3}{5} - \frac{2}{5} \log_2 \frac{2}{5} = 0.97$$

$$S_{Strong} \leftarrow \{0+, 2-\} \quad Entropy(S_{Strong}) = 0.0$$

$$S_{Weak} \leftarrow \{3+, 0-\} \quad Entropy(S_{Weak}) = 0.0$$

$$Gain(S_{Rain}, Wind) = Entropy(S) - \sum_{v \in \{Strong, Weak\}} \frac{|S_v|}{|S|} Entropy(S_v)$$

$$Gain(S_{Rain}, Wind) = Entropy(S) - \frac{2}{5} Entropy(S_{Strong}) - \frac{3}{5} Entropy(S_{Weak})$$

$$Gain(S_{Rain}, Wind) = 0.97 - \frac{2}{5} \cdot 0.0 - \frac{3}{5} \cdot 0.0 = 0.97$$

Information Gain & Conclusions

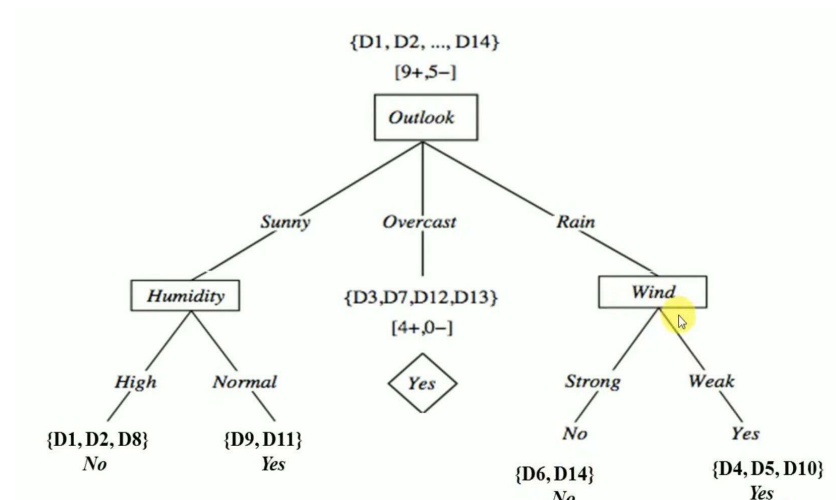
- Wind has highest gain and taken as decision node
 - has two possibilities like Weak & Strong
- If Wind == Weak the target (Play Tennis) = Yes & Wind == Strong the target (play Tennis) = No

Day	Temp	Humidity	Wind	Play Tennis
D4	Mild	High	Weak	Yes
D5	Cool	Normal	Weak	Yes
D6	Cool	Normal	Strong	No
D10	Mild	Normal	Weak	Yes
D14	Mild	High	Strong	No

$$Gain(S_{Rain}, Temp) = 0.0192$$

$$Gain(S_{Rain}, Humidity) = 0.0192$$

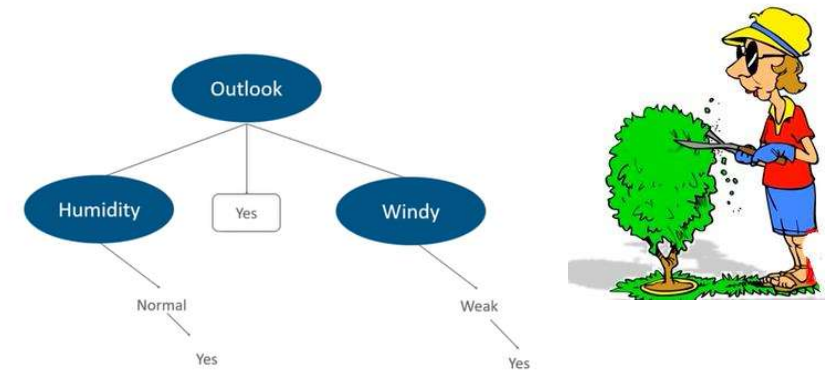
$$Gain(S_{Rain}, Wind) = 0.97$$



Solve the Example for constructing the decision tree (use classification as target variable)

Instance	Classification	a1	a2
1	+	T	T
2	+	T	T
3	-	T	F
4	+	F	F
5	-	F	T
6	-	F	T

Pruning : optimizing the decision tree



Removing the unnecessary branches or paths that will not trace through the tree for any decisions

Are tree based models better than linear models

- If the relation between independent and dependent is well approximated by linear then go with linear regression then decision tree.
- If the data is highly non linear and complex decision tree outperforms better than regression methods.
- To represent the data is easy form to understand.

Machine Learning Methods
(Applications of models)

3. Unsupervised Machine Learning

What is Unsupervised learning (Example of a cricket playing video)

- 2 teams , India vs Australia
- Different kinds of players, such as batsmen, bowlers and so on.
- Ball hits wickets or is caught, batsman is out
Cheer when India had 4 or 6.



Why it is important

- Find patterns which are previously unknown
- Patterns help in categorizing or finding association
- Detect anomalies and defects in the data.
- Work on unlabelled data

Types of understanding from Unsupervised Learning

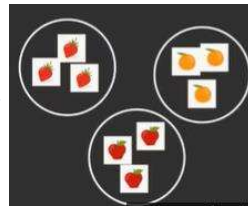
- Clustering
 - Patterns to find the data like, shape, colour
 - K-Means, KNN, hierarchical



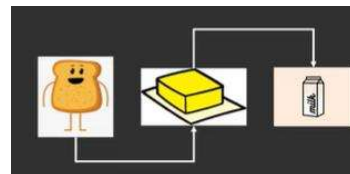
Analyse
the data



Categorize
based on
shape, size
and colour



- Association
 - Finding the association of one data item to another data item and map them to analyse better predictions
 - Apriori and FP growth
 - Example : Super market analysis
 - Mapping the items of bread with butter and milk



K-Means Clustering

- What is clustering
- Types of clustering
- What is K-Means clustering
- How does a K-Means algorithm works?

What is clustering

“Clustering is the process of dividing the datasets into groups, consisting of similar data-points”

- Points in the same group are as similar as possible
- Points in different group are as dissimilar as possible

Items arranged in a mall

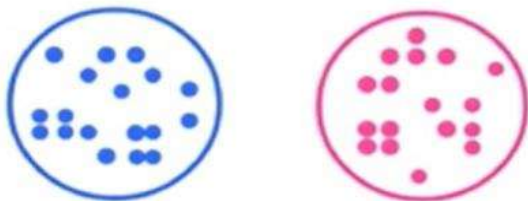


Types of Clustering

- Exclusive clustering
- Overlapping clustering
- Hierarchical clustering

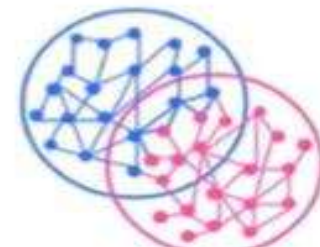
Exclusive clustering

- Hard Clustering
- Data Point / Item belongs exclusively to one cluster
- For Example: K-Means Clustering

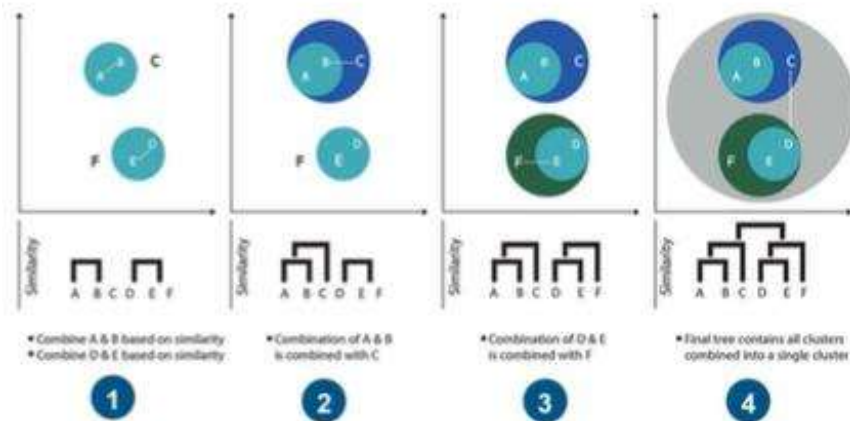


Overlapping clustering

- Soft Cluster
- Data Point/ Item belongs to multiple cluster
- For Example: Fuzzy/ C-Means Clustering



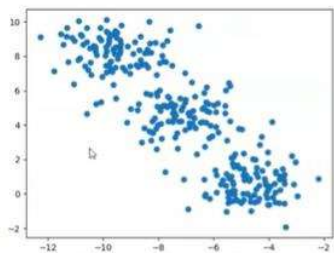
Hierarchical clustering



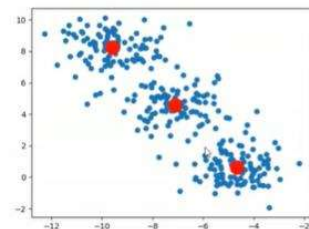
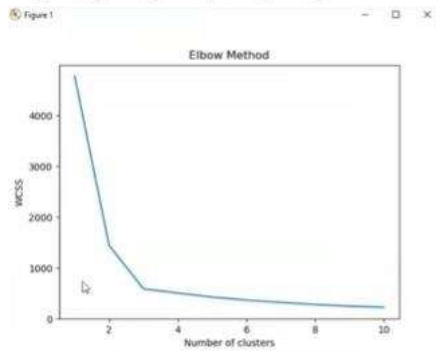
What is K-means clustering

- Identifying the data points to which cluster (out of K clusters) it belongs to.
- Centroid is the head for each cluster
- Example : Laundering the cloths for washing
 - Vacate the hostel
 - Have huge cloths to sent for laundry
 - Mother asked you keep three groups all cloths
 - You will order those groups based on temperature of water required, time required and type of washing required.

Elbow methods (K-means Clustering)



- Suggest appropriate number of clusters to optimize the data set.



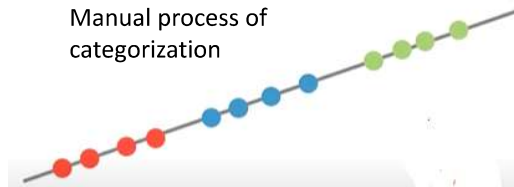
Steps in K-Means Algorithm (for one iteration)

1. Select the number of clusters to be identified, i.e. $K=3$ here .
2. Randomly select 3 distinct data point as centroids.
3. Measure the distance between the 1st point and selected 3 clusters.
4. Assign the 1st point and to nearest cluster
 1. Perform 3 & 4 steps with respect to all data points
5. Check the variance of each iteration
6. Continue step 1 to

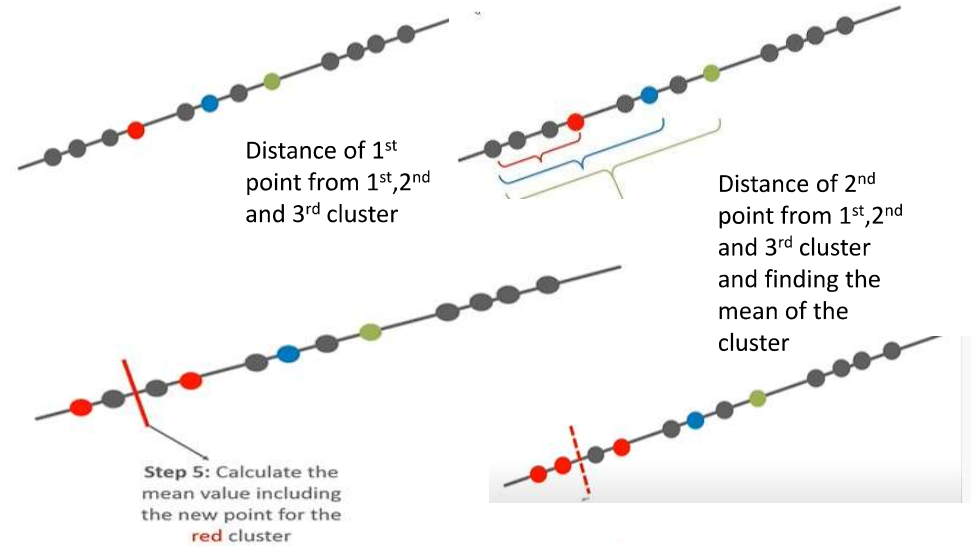
Example to Solve



Manual process of categorization

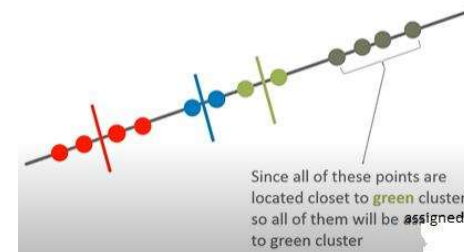


Various steps



Which point belongs to which cluster?

Results

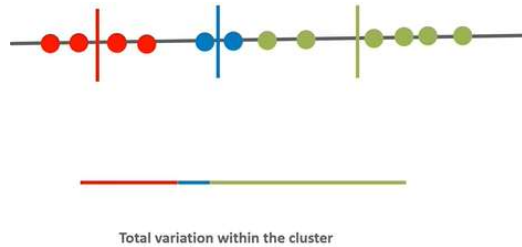


Not accurate results



Variance in data points

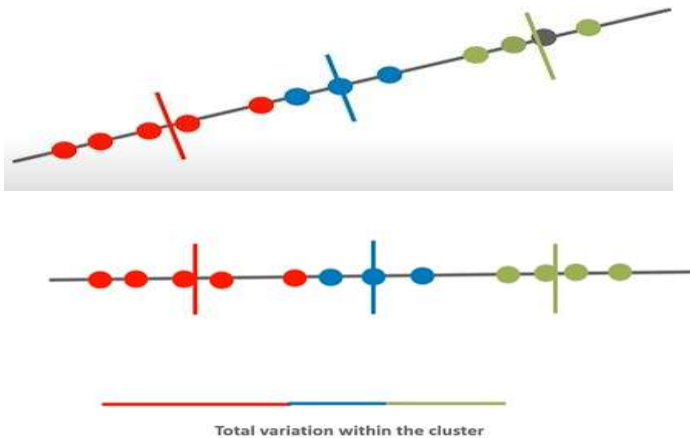
- Variation of the data points in a cluster



- How to improve?
 - K-Means performs iteration again and again unless and until the data points within each cluster stop changing.
 - Iteration with different random points to cluster heads.

Iteration 2

Algorithm picks 3 initial clusters and adds the remaining points to the cluster with the nearest mean, and again recalculating the mean each time a new point is added to the cluster.



Iteration 2

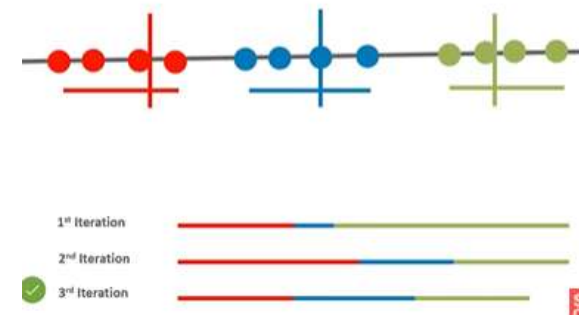
Iteration 2: Again we will start from the beginning. But this time we will be selecting different initial random point (as compared to what we chose in the 1st iteration)



- **Step 1:** Select the number of clusters to be identified, i.e. $K=3$ in this case
- **Step 2:** Randomly select 3 distinct data point
- **Step 3:** Measure the distance between the 1st point and selected 3 clusters

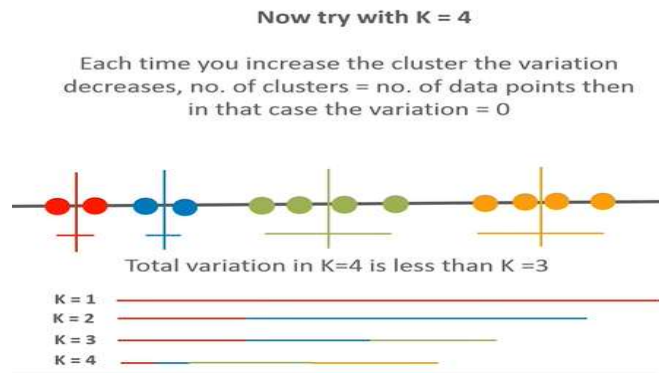
Results for Three Iterations

The algorithm can now compare the result and select the best variance out of it

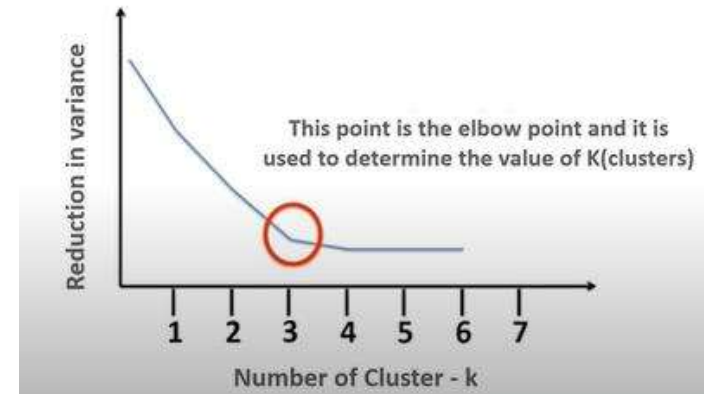


How to Chose K

- Number of iterations repeated , after some time the variance will not effected.
- How many iterations can be decided.
- Trail and error method.
- K =1 is worst case scenario.



Variance vs K



K-Means Clustering using L1 Distance

- L1 distance = the sum of all absolute differences;; $|x_2 - x_1| + |y_2 - y_1|$

- L1 distance is just manhattan distance: sum of differences in each dimension – **ITERATION 1**

Data Point	C1: (1, 0, 0)	C2: (0, 1, 1)	Cluster
P1: (1, 2, 3)	$0+2+3 = 5$ ✓	4	C2
P2: (0, 1, 2)	$1+1 = 2$	1	C2
P3: (3, 0, 5)	7	8	C1
P4: (4, 1, 3)	7	6	C2
P5: (5, 0, 1)	5	6	C1

Finding new centroids using averages

- Average all newly formed clusters ,P1, P2 & P4 belongs to C2 so its new centroid is::
 $(1+0+4/3, 2+1+1/3, 3+2+3/3)=(1.16,1.33,2.66)$
- P3 & P4 belongs to C1 so the new centroid is ::
 $(3+5/2,0+0/2,5+1/2) = (4,0,3)$

With new centroids

- L1 distance is just manhattan distance: sum of differences in each dimension - **ITERATION 2**

Data Point	C1: (4, 0, 3)	C2: (1.6, 1.3, 2.6)	Cluster
P1: (1, 2, 3)	5	1.7	C2
P2: (0, 1, 2)	6	2.5	C2
P3: (3, 0, 5)	3	5.3	C1
P4: (4, 1, 3)	1	3.1	C1
P5: (5, 0, 1)	3	6.3	C1

When to stop the iterations

- Continued until no movement of data points to new clusters compared to previous clusters

- L1 distance is just manhattan distance: sum of differences in each dimension - **ITERATION 3**

Data Point	C1: (4, 0.33, 3)	C2: (0.5, 1.5, 2.5)	Cluster
P1: (1, 2, 3)	4.67	1.5	C2 ✓
P2: (0, 1, 2)	5.67	1.5	C2 ✓
P3: (3, 0, 5)	3.33	6.5	C1
P4: (4, 1, 3)	0.67	4.5	C1
P5: (5, 0, 1)	3.33	7.5	C1

The cluster points are same as the previous cluster formation, stopped the iterations

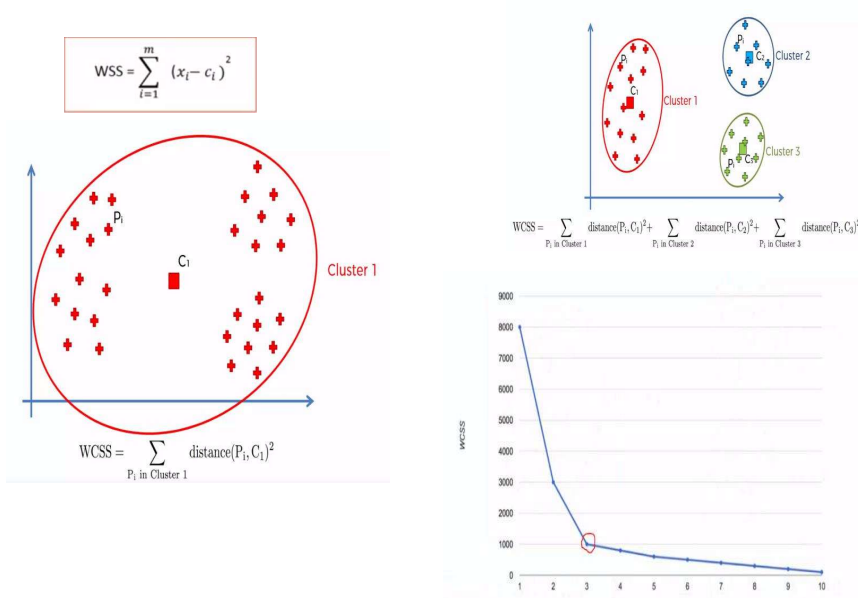
Disadvantages in K-Means

- Identifying the initial centroids
- Specifying the number of clusters for the given dataset in advance.

Optimal methods for K selection

- Elbow Method
- Silhouette Method

WCSS(Within Cluster Sum of Squares)



K-Means Clustering-Elbow Method

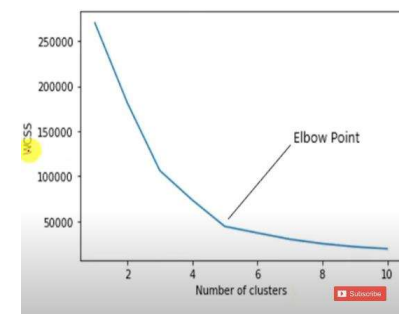
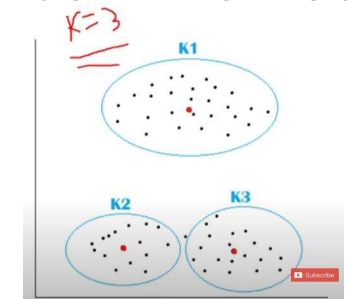
- **Step 1:** Apply K-Means clustering algorithm and form the clusters for different values of k. For example, k=[1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
- **Step 2:** For each k calculate the within-cluster sum of squares (WCSS).

$$WCSS = \sum_k \left(\sum_{d_i \text{ in } C_k} \text{distance}(d_i, C_k)^2 \right)$$

Where,

C is the cluster centroids and d is the data point in each Cluster.

- **Step 3:** Plot curve of WCSS according to the number of clusters.
- **Step 4:** The location of bend in the plot is generally considered an indicator of the approximate number of clusters.



Applications

- Recognize patterns to cluster data
- Defects in the data collected.
- Recommendation systems for various business models in amazon, required movies based on your patterns of watching movies
- Retails about customer behaviours
- Image segmentation webpage segmentation, data processing
- Insurance companies: fraud detection, risk factor identification



Amazon learns the customer's purchase and recommend the products which are most frequently bought together.



Various patterns of the user and their usage of the credit card are studied by the algorithm. If the card is used in ways that do not match the behavior, an alarm is generated possibly meaning fraud.

Disadvantages of unsupervised ML

- Less accurate
- Output obtained may not match with the actual data.
- Output need to be understood by the user to map the labels

Supervised vs Un supervised learning

Parameter	Supervised Learning	Unsupervised Learning
Dataset	Labelled Dataset	Unlabeled Dataset
Method of Learning	Guided Learning	Algorithm learns by itself using dataset
Complexity	Simpler Method	Computationally Complex
Accuracy	More accurate	Less accurate

References

- <https://www.youtube.com/watch?v=qDcl-FRnwSU>
- <https://www.youtube.com/watch?v=UhVn2WrzMnI>
- <https://www.youtube.com/watch?v=1XqG0kaJVHY>
- <https://towardsdatascience.com/machine-learning-algorithms-part-9-k-means-example-in-python-f2ad05ed5203>.
- <https://online.hbs.edu/blog/post/data-life-cycle>.
- <https://pressbooks.bccampus.ca/iotbook/chapter/iot-use-cases/>

Benefits of Reusing IoT Solutions

Reusing IoT Solutions & IoT reusability strategy/plan/steps

- **Cost Savings:** Avoids the high costs of developing hardware and software from scratch.
- **Faster Deployment:** Speeds up time to market by using pre-built components.
- **Proven Reliability:** Reduces risks by leveraging tested solutions.
- **Interoperability:** Many existing IoT solutions support industry standards, making integration easier.

Areas Where IoT Solutions Can Be Reused

- **Hardware:**
 - Sensors (temperature, humidity, motion, etc.)
 - Microcontrollers and single-board computers (Arduino, Raspberry Pi, ESP8266, ESP32)
 - Gateways and connectivity modules (LoRa, Zigbee, Wi-Fi, Bluetooth)
- **Software:**
 - IoT platforms (AWS IoT, Azure IoT, Google Cloud IoT, ThingsBoard)
 - Open-source firmware (Tasmota, ESPHome, OpenWrt)
 - APIs and SDKs for common tasks (MQTT, CoAP, HTTP)
- **Network and Protocols:**
 - Reusing MQTT brokers (Eclipse Mosquitto, HiveMQ)
 - Existing LPWAN solutions (LoRaWAN, NB-IoT, Sigfox)
- **Cloud and Edge Computing:**
 - Serverless functions for IoT data processing
 - AI/ML models for predictive maintenance using edge AI solutions

Challenges and Considerations

- **Compatibility:** Ensuring reused components work together.
- **Security Risks:** Older solutions may have vulnerabilities; updating firmware and using encryption is essential.
- **Scalability:** Some existing solutions may not handle future expansion needs.
- **Licensing and Compliance:** Ensure reused software and hardware comply with industry regulations.

Best Practices

- **Adopt Modular Design:** Use a **flexible architecture** that allows swapping components when needed.
- **Utilize Open Standards:** Choose IoT solutions that **support widely accepted** protocols (MQTT, CoAP).
- **Regular Updates:** Keep **firmware and security patches up to date**.
- **Use Cloud Services:** Many **cloud providers offer scalable IoT solutions** with built-in security and analytics.

IoT Solution Reusability Plan

maximize efficiency, reduce costs,
and enhance scalability.

1. Define Reusability Objectives

- **Reduce Development Time & Cost:** Minimize new investments by leveraging existing hardware/software.
- **Enhance Interoperability:** Ensure compatibility between legacy and new IoT components.
- **Ensure Scalability & Future-Proofing:** Select solutions that can grow with business needs.
- **Maintain Security & Compliance:** Keep reused solutions up to date with security patches.

2. Identify Reusable IoT Components

- **Hardware**
 - **Sensors & Actuators:** Temperature, humidity, motion, gas, pressure sensors, etc.
 - **Microcontrollers & SBCs:** ESP32, Raspberry Pi, Arduino, STM32, etc.
 - **Connectivity Modules:** LoRa, Zigbee, Bluetooth, Wi-Fi, NB-IoT, LTE-M.
 - **Gateways & Edge Devices:** Edge AI devices, Industrial IoT gateways.

2. Identify Reusable IoT Components

- **SofIoT Platforms:** AWS IoT Core, Azure IoT Hub, Google Cloud IoT, ThingsBoard.
- **Middleware & Protocols:** MQTT, CoAP, WebSockets, OPC UA.
- **Open-Source Software:** Tasmota, ESPHome, Node-RED, OpenHAB.
- **APIs & SDKs:** Pre-built libraries for IoT data handling and integration.
- **Middleware & Platforms**

4. Data & Analytics

- **AI & ML Models:** Predictive maintenance, anomaly detection, image recognition models.
- **Data Storage Solutions:** Cloud databases, edge storage, blockchain for IoT.

3. Network & Security

- **Pre-configured Network Architectures:** LoRaWAN, 5G, LPWAN-based solutions.
- **Security Modules:** TLS encryption, VPN solutions, hardware security chips.

5. Standardize and Modularize

- **Use Open Standards:** Ensure all reused components follow industry standards (e.g., MQTT, OPC UA).
- **Implement Modular Design:** Break down IoT solutions into reusable modules (e.g., device firmware, analytics engine).
- **Ensure API-First Development:** Enable easy integration with third-party services.

6. Assessment & Validation

- Hardware Compatibility Testing:** Check power, communication interfaces, and environmental factors.
- Software & Protocol Validation:** Ensure APIs, SDKs, and protocols function correctly.
- Security & Compliance Audit:** Perform penetration testing and update outdated security measures.
- Performance Benchmarking:** Test scalability and efficiency under different workloads.

8. Continuous Improvement & Lifecycle Management

- Regularly Update & Patch Reused Components:** Ensure ongoing security and reliability.
- Monitor & Optimize Performance:** Collect usage data for improving efficiency.
- Plan for Future Upgrades:** Keep reusability in mind for future expansions.

7. Documentation & Knowledge Sharing

- Create a Centralized Repository:** Maintain a database of reusable IoT assets.
- Provide Usage Guidelines:** Standard operating procedures (SOPs) for integration.
- Train Teams:** Conduct training on how to effectively reuse IoT components.

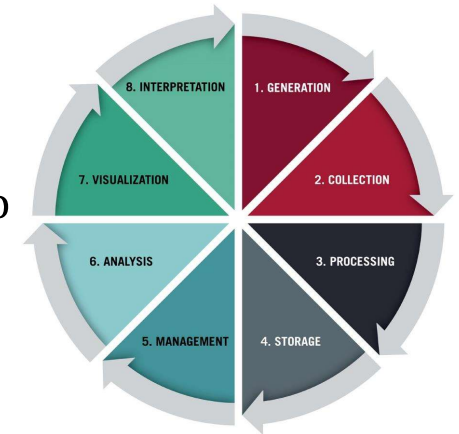
9. Implementation & Deployment Strategy

- Pilot Testing:** Run a small-scale deployment before full rollout.
- Gradual Integration:** Phase out legacy solutions while reusing key components.
- Feedback Mechanism:** Continuously gather insights from end-users for improvements.

8 STEPS IN THE DATA LIFE CYCLE

Stages of data lifecycle

- Data LifeCycle Management is a process that helps organizations to manage the flow of data throughout its lifecycle – from initial creation through to destruction or deployment.



Generation

- For the data life cycle to begin, data must first be generated.
 - Every sale, purchase, hire, communication, interaction—everything generates data.
- Given the proper attention, this data can often lead to **powerful insights** that allow you to better serve your customers and become more effective in your role.
- Data is typically generated by an organization in one of 3 ways:
 - **Data Acquisition:** acquiring already existing data which has been produced outside the **organization**
 - **Data Entry:** **manual entry** of new data by personnel within the organization
 - **Data Capture:** capture of data generated by devices used in various processes **in the organization**

Collection

- Not all of the data that's generated every day collected is used.
- It's up to your data team to identify what information should be captured and the best means for doing so, and what data is unnecessary or irrelevant to the project at hand.
- Data can be collected in variety of ways, including:
 - **Forms:** Web forms, client or customer intake forms, vendor forms, and human resources applications are some of the most common ways businesses generate data.
 - **Surveys:** Surveys can be an effective way to gather vast amounts of information from a large number of respondents.
 - **Interviews:** Interviews and focus groups conducted with customers, users, or job applicants offer opportunities to gather qualitative and subjective data that may be difficult to capture through other means.
 - **Direct Observation:** Observing how a customer interacts with your website, application, or product can be an effective way to gather data that may not be offered through the methods above.

Processing

- Once data has been collected, it must be processed. Data processing can refer to various activities, including:
 - **Data wrangling, Data cleaning**, in which a data set is cleaned and transformed from its raw form into something more accessible and usable. This is also known as data cleaning, data munging, or data remediation.
 - **Data compression, Data Transformation** in which data is transformed into a format that can be more efficiently stored.
 - **Data encryption, Data reduction** in which data is translated into another form of code to protect it from privacy concerns.
- Even the simple act of taking a printed form and digitizing it can be considered a form of data processing.

Management

- Data management, also called database management, involves organizing, storing, and **retrieving data as necessary over the life of a data project.**
- While referred to here as a “step,” it’s **an ongoing process that takes place from the beginning through the end of a project.**
- Data management includes everything **from storage and encryption to implementing access logs and changelogs that track who has accessed data** and what changes they may have made.

Storage

- After data has been collected and processed, it must be stored for **future use.**
- This is most commonly **achieved through the creation of databases or datasets.**
- These datasets may then be **stored in the cloud**, on servers, or using another **form of physical storage** like a hard drive, CD, cassette, or floppy disk.
- When determining how to **best store** data for your organization, it’s important to build in a certain level of redundancy **to ensure that a copy of data will be protected and accessible, even if the original source becomes corrupted or compromised.**

Analysis

- Data analysis refers to processes that attempt to get **meaningful insights** from raw data.
- Analysts and data scientists **use different tools and strategies to conduct these analyses.**
- Some of the more commonly used methods include **statistical modeling, algorithms, artificial intelligence, data mining, and machine learning.**
- Exactly who performs an analysis depends on the specific challenge being addressed, as well as the size of your organization’s data team.
 - Frame questions for your requirement
- **Business analysts, data analysts or engineer, and data scientists can all play a role.**

Visualization

- Data visualization refers to the process of creating **graphical representations of your information**, typically through the use of one or more visualization tools.
- Visualizing data makes **it easier to quickly communicate your analysis to a wider audience both inside and outside** your organization.
- The form visualization takes depends on the data working with, as well as **the story you want to communicate**.
- While technically not a required step for all data projects, data visualization has become an increasingly important part of the data life cycle.

DATA VISUALIZATION TOOLS FOR BUSINESS

- Microsoft Excel (and Power BI)
- Google Charts
- Tableau
- Zoho Analytics
- Datawrapper
- Infogram

Data Visualization

- creating a visual representation of the information.
- most common [data visualization techniques](#) include:
 - Pie charts
 - Bar charts
 - Histograms
 - Gantt charts
 - Heat maps
 - Box-and-whisker plots
 - Waterfall charts
 - Area charts
 - Scatter plots
 - Infographics
 - Maps

Interpretation

- Finally, the interpretation phase of the data life cycle provides the opportunity to make **sense of your analysis and visualization**.
- Beyond simply presenting the data, this is when you **investigate it through the lens** of your expertise and understanding.
- Interpretation may not only include a **description or explanation of what the data shows but**, more importantly, what the implications may be.